EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

Name: *Kanishka S* | ***Register Number:*** *240701227*

Subject: *Low Code Application Development using Zoho Creator*

Aim

To implement and explore the security, access-control, data-sharing and governance features of the Smart Campus Sustainability & Carbon Tracker application in Zoho Creator.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use. On a campus, for example, a data entry staff member needs to see the complaints handled by the facility team but does not need to change the resolution details.

Application Used

The practical was carried out in Zoho Creator, where the security and governance features of the application editor were explored. The Smart Campus Sustainability & Carbon Tracker application manages campus energy, water, waste, activities and complaints, and the settings studied here (User Permissions, Roles and Data Sharing) are available in every Zoho Creator application.

Security and Governance Architecture

Figure 1 is a conceptual supporting diagram of the security and governance structure of the Smart Campus application. It is not a screenshot of the Zoho Creator interface.

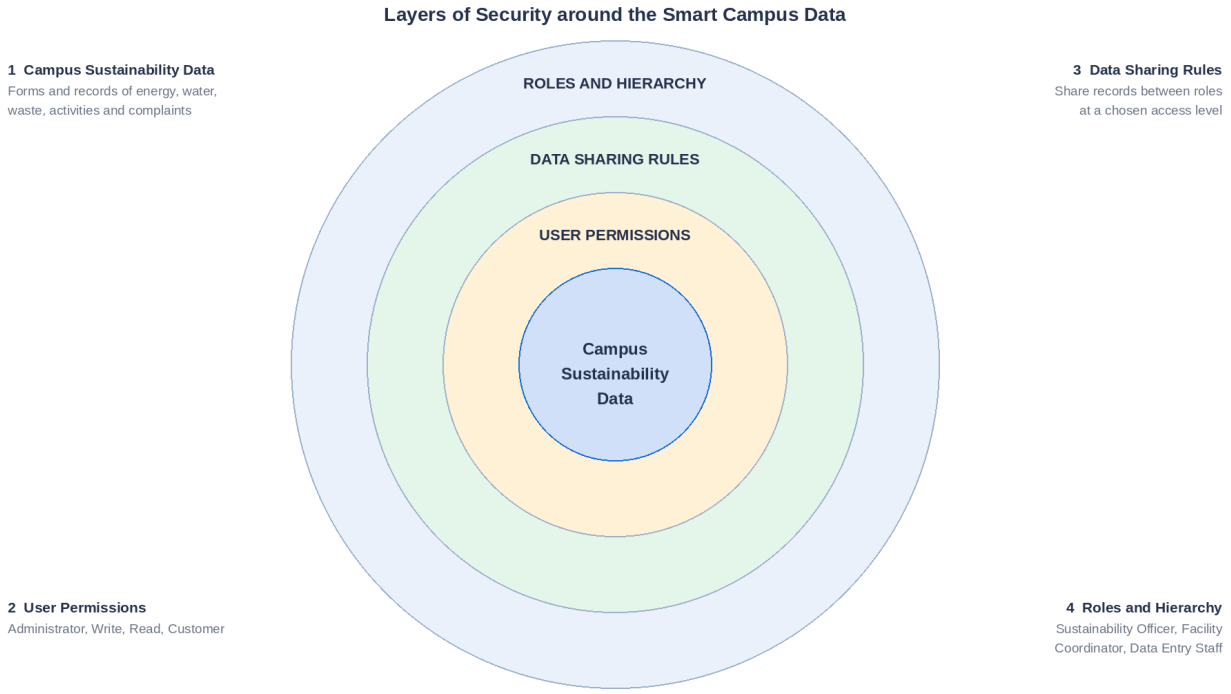


Figure 1: Security & Governance Layers (Conceptual Diagram)

The diagram shows the campus sustainability data of the application at the core, protected by successive layers. User permissions define what a user may do. Data sharing rules extend the visibility of records between roles or users at a chosen level of access. The role hierarchy forms the outermost layer and influences which records each role can see. The key outcomes are controlled access to data, role-based visibility, safe collaboration through data sharing, and efficient administration.

User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records for each user. The Permissions tab lists the permission sets available in the application and provides an Add Permission option. The application listed four permissions:

Name	Description (as displayed)
Administrator	This profile will have full access to all components
Write	This profile will have write permission for all components
Read	This profile will have read permission for all components
Customer	This profile can add and view its own records

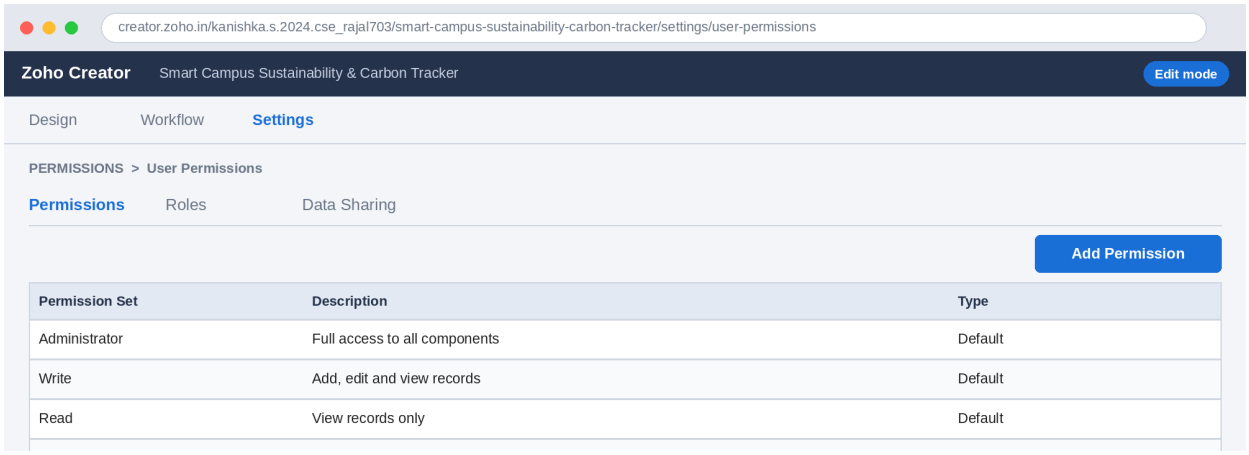


Figure 2: User Permissions (Permissions tab) in the Zoho Creator application editor (illustration)

Read permission allows a user to view the components and records of the application without changing them, whereas Write permission allows the user to make changes, such as adding or modifying records. Further profiles, for example one for the facility coordinators, can be created with the Add Permission option; none was added in this experiment.

Role-Based Access Control

The Roles tab shows the role hierarchy of the application. The tree contains the default top-level role, CEO, and the roles created in Experiment 8 beneath it. The options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available for building the hierarchy.

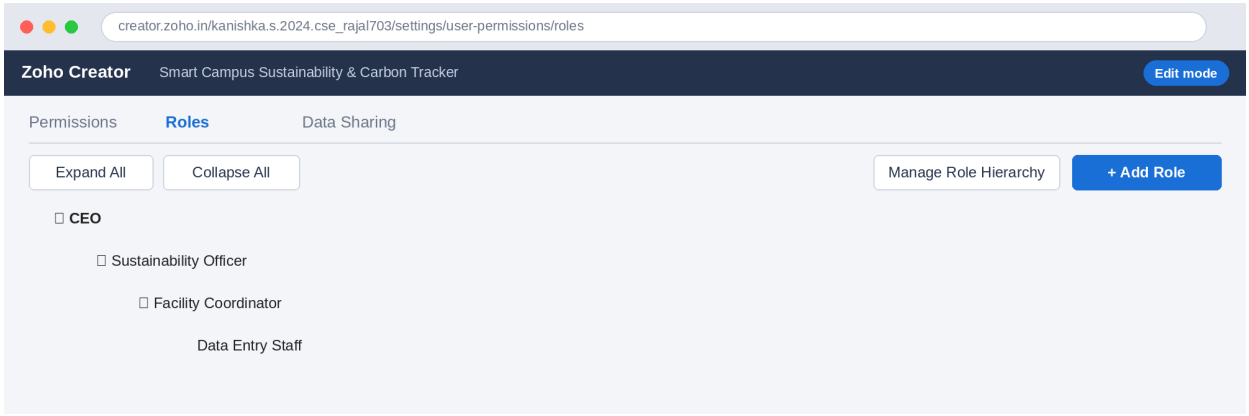


Figure 3: Roles and Role Hierarchy (Roles tab) in the Zoho Creator application editor (illustration)

For the Smart Campus application, the following role hierarchy is used.

Role	Reports To	Position in Hierarchy
Sustainability Officer	CEO	Top level of the campus roles
Facility Coordinator	Sustainability Officer	Second level
Data Entry Staff	Facility Coordinator	Third level (lowest role)

Arranging roles in this way gives the application a clear reporting structure. The position of a role in the hierarchy influences which records its members can see, and further access can be given to a specific role through data sharing rules, as described in the next section.

Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user from which records are shared, the role or user to which they are shared, and the level of access given. In this way, records can be made visible to additional roles or users beyond the default visibility of the role hierarchy. The data sharing rule configured for the application has the following details:

Parameter	Value
Rule Name	Sustainability_Complaint_Form
Form	Sustainability Complaint
Records Shared From	Roles – Facility Coordinator
Records Shared To	Roles – Data Entry Staff
Level of Access	Read Only

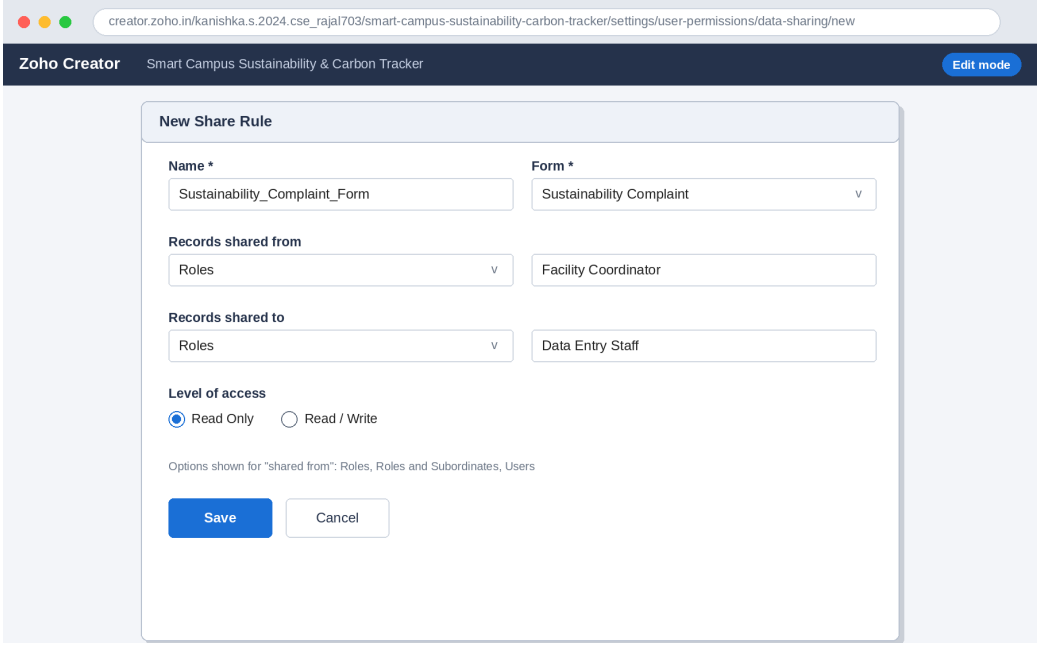


Figure 4: New Share Rule panel (Data Sharing tab) in the Zoho Creator application editor (illustration)

Figure 4 shows the New Share Rule panel opened from the Data Sharing tab. The panel has the fields Name, Form, Records shared from (with a role selector), Records shared To (with a role selector) and Level of access, which is Read Only by default. The rule was saved and appears in the Data Sharing list.

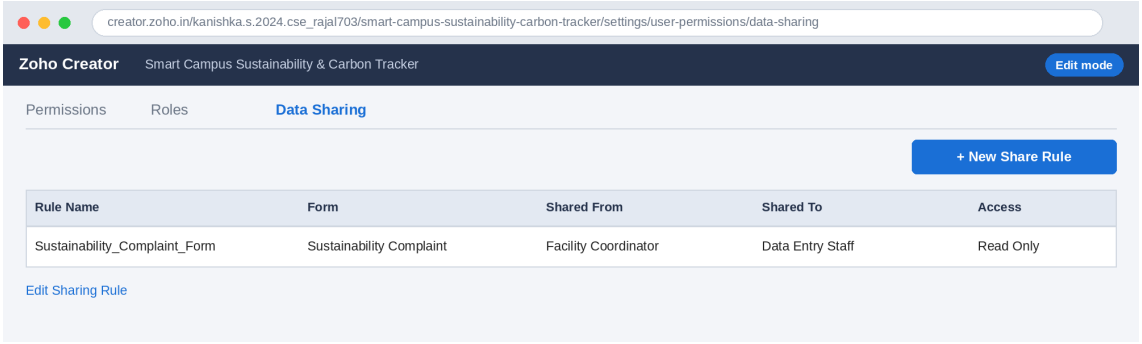


Figure 4a: Configured rule in the Data Sharing list (illustration)

The Records shared from field offers three choices: Roles, Roles and Subordinates, and Users. Roles shares the records of a single selected role, Roles and Subordinates shares the records of a role together with the roles beneath it in the hierarchy, and Users shares the records of selected individual users. With

the configured rule, Roles is selected, with the Facility Coordinator role as the source; with Read Only access, members of the Data Entry Staff role can view the shared records of the complaint form but cannot modify them.

Security Features and Governance Controls

Figure 5 is a conceptual diagram that classifies the security and governance controls of Zoho Creator as they apply to the Smart Campus application. It is not a screenshot, and it is not proof that every listed feature was configured.

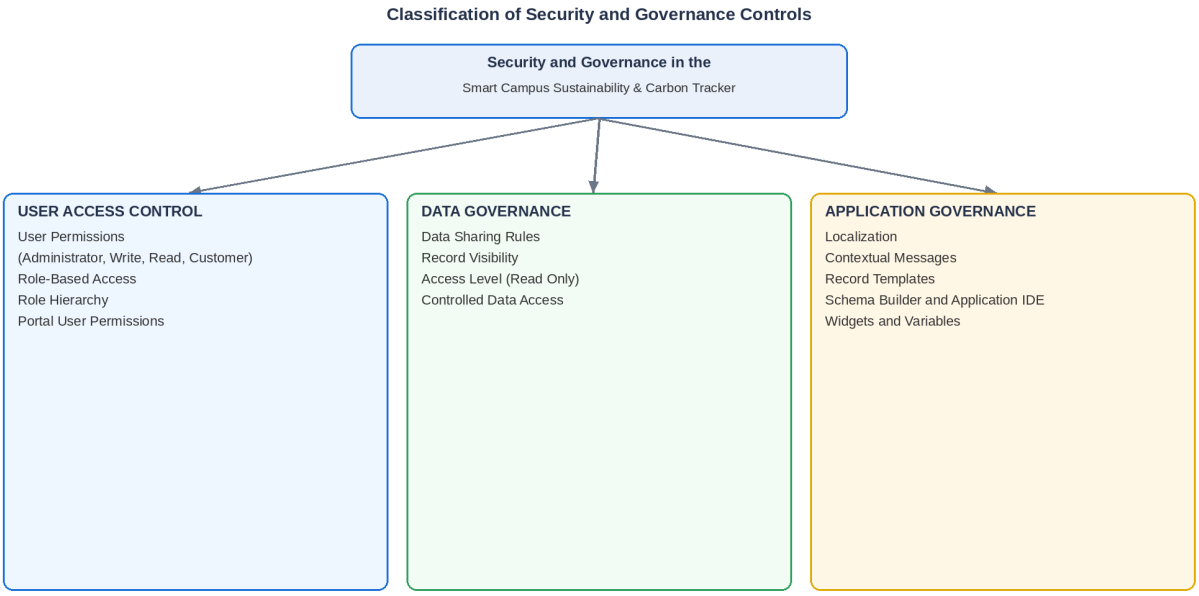


Figure 5: Classification of Security and Governance Controls (Conceptual Diagram)

Control	Explanation
User access control	User permissions, role-based access and portal user permissions determine who can use the application and what they can do.
Data governance	Data sharing, access level, record visibility and role hierarchy govern how application data is accessed.
Application governance	Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application.
Role hierarchy	Arranges roles in a reporting structure that influences which records a role can see.
Data sharing	Extends the visibility of records from one role or user to another with a chosen access level.
Record visibility	Decides which records are visible to which users, based on roles and sharing rules.
Administrative controls	Allow an administrator to manage permissions, roles and application settings.

In this practical, user permissions, roles and the data sharing rule were examined and configured. The other items in the diagram, such as portal user permissions, localization, schema builder and widgets, are available in the Settings menu of the application, but they were not configured in this experiment.

Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users. It was not configured in this experiment, and no portal permissions were set for the Smart Campus application.

Role-Based Access and Data Sharing Flow

Figure 6 is a conceptual diagram that combines the role hierarchy of the application with the data sharing rule. On the left, the roles Sustainability Officer, Facility Coordinator and Data Entry Staff are shown in reporting order. On the right, the rule Sustainability_Complaint_Form is traced from the complaint form, through the role it is shared from, to the role it is shared to and the level of access given.

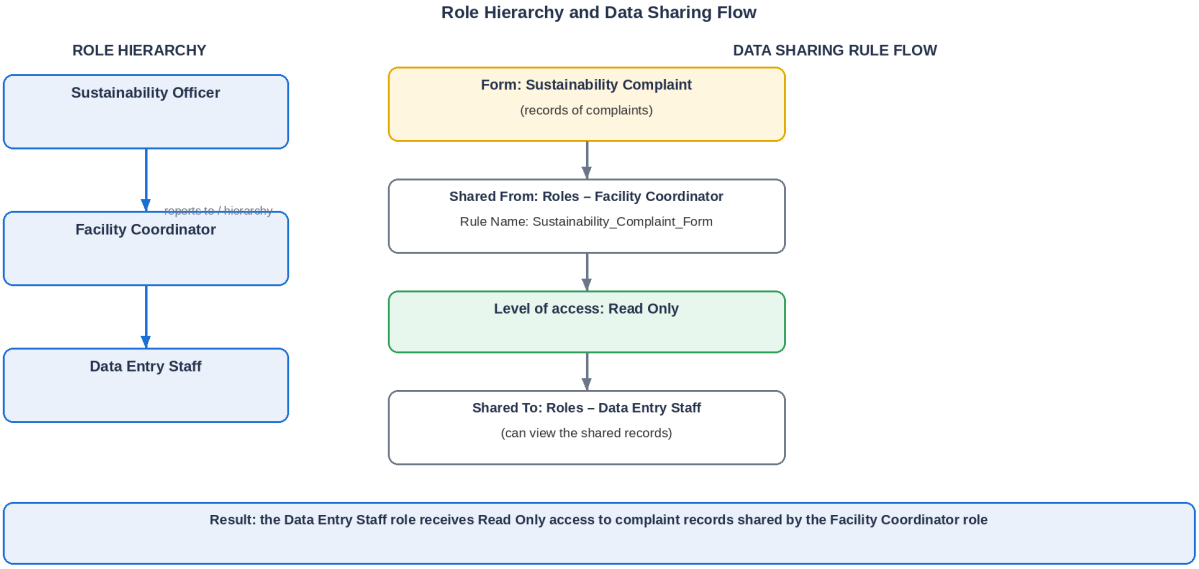


Figure 6: Role Hierarchy and Data Sharing Flow (Conceptual Diagram)

- The roles of the application form a hierarchy: Sustainability Officer, Facility Coordinator and Data Entry Staff.
- The data sharing rule Sustainability_Complaint_Form applies to the complaint form.
- Records are shared from the Facility Coordinator role to the Data Entry Staff role.
- The level of access is Read Only, so the Data Entry Staff role can view the shared records but not modify them.

Governance Observations

- Predefined permission profiles (Administrator, Write, Read and Customer) were available in the application.
- The Roles tab showed the default CEO role with the three campus roles beneath it.
- The Data Sharing tab lists the rule Sustainability_Complaint_Form, which shares records from the Facility Coordinator role to the Data Entry Staff role with Read Only access.
- Portal User Permissions was not configured in this experiment.
- The practical showed how access can be limited to what a role needs by choosing a suitable level of access.

Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the permission profiles were examined, the role tree of the application editor was viewed, and the New Share Rule panel of the Data Sharing tab was used. A role hierarchy of Sustainability Officer, Facility Coordinator and Data Entry Staff and a data sharing rule named Sustainability_Complaint_Form, sharing complaint records from the Facility Coordinator role to the Data Entry Staff role with Read Only access, were applied to the Smart Campus application.

The experiment shows how permissions, roles and data sharing work together so that each role sees the records it needs and no more. Choosing a suitable level of access, such as Read Only, allows records to be shared for collaboration without allowing them to be changed. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications.